ASSIGNMENT-9.5

NAME: ARCHITHA

ROLL NO: 2303A51001

BATCH: 30

PROBLEM-1

PROMPT:

def reverse_string(text):

return text[::-1]

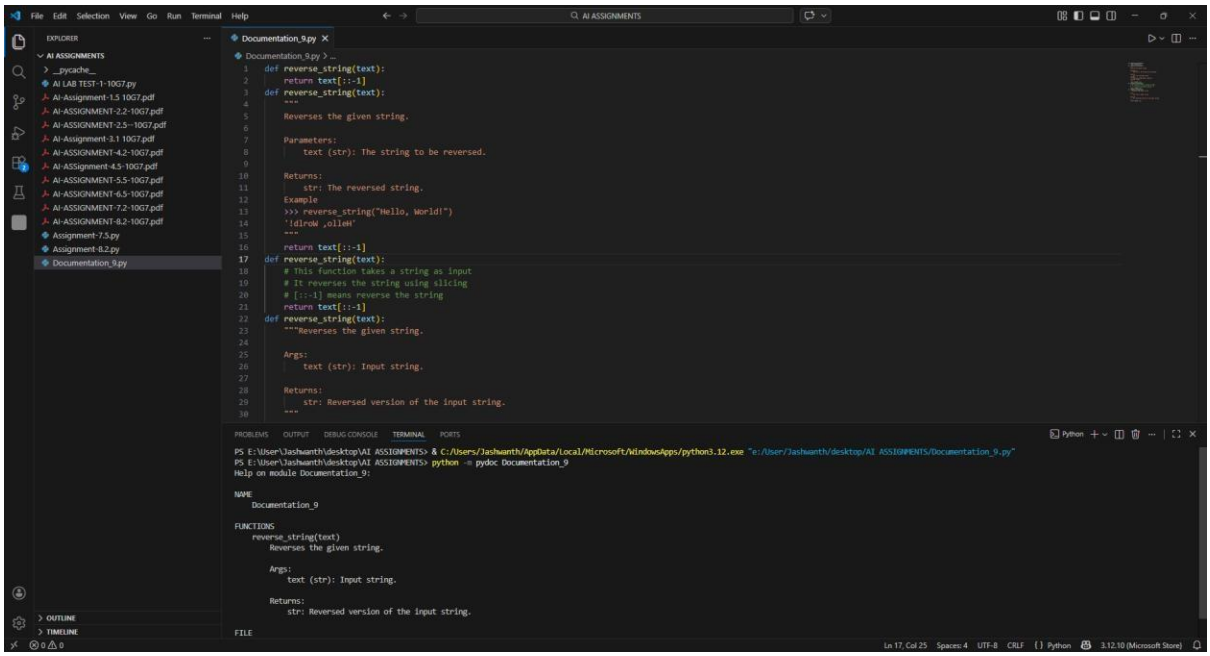
Write documentation in:

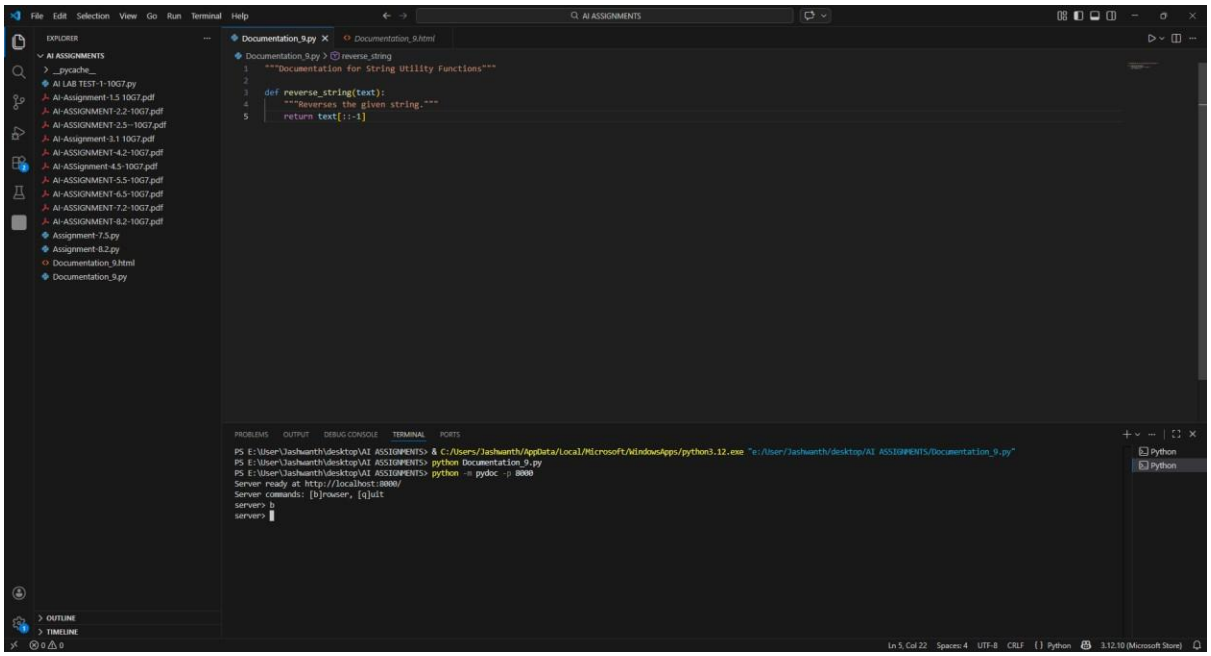
(a) Docstring

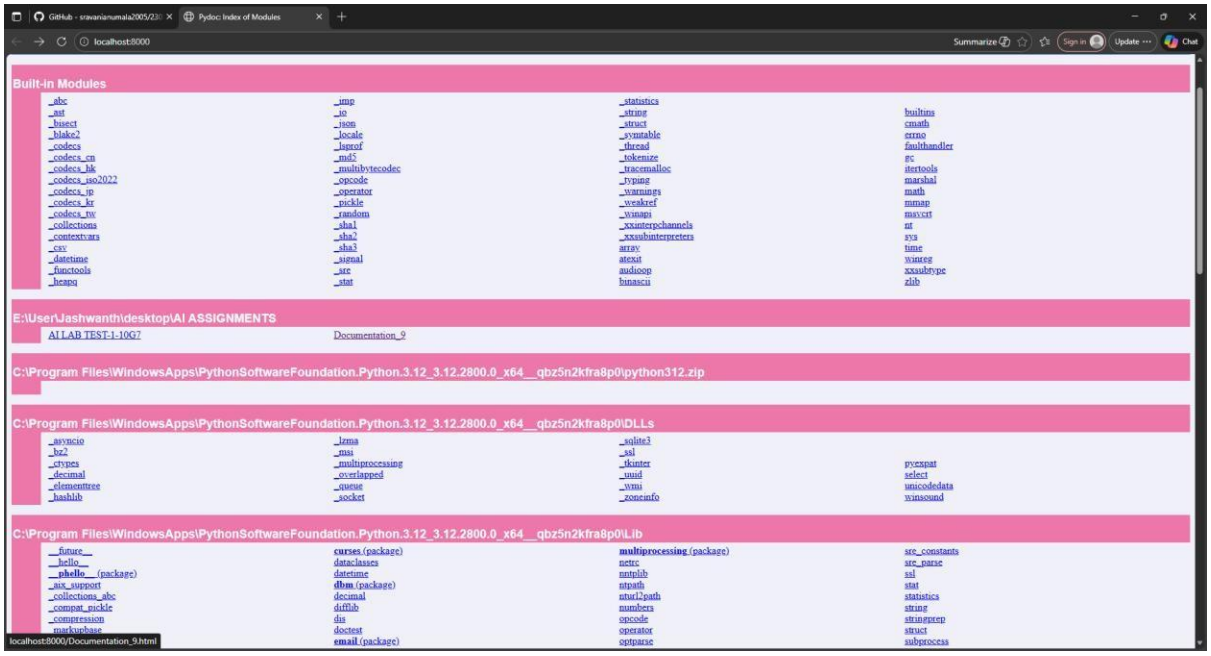
(b) Inline comments

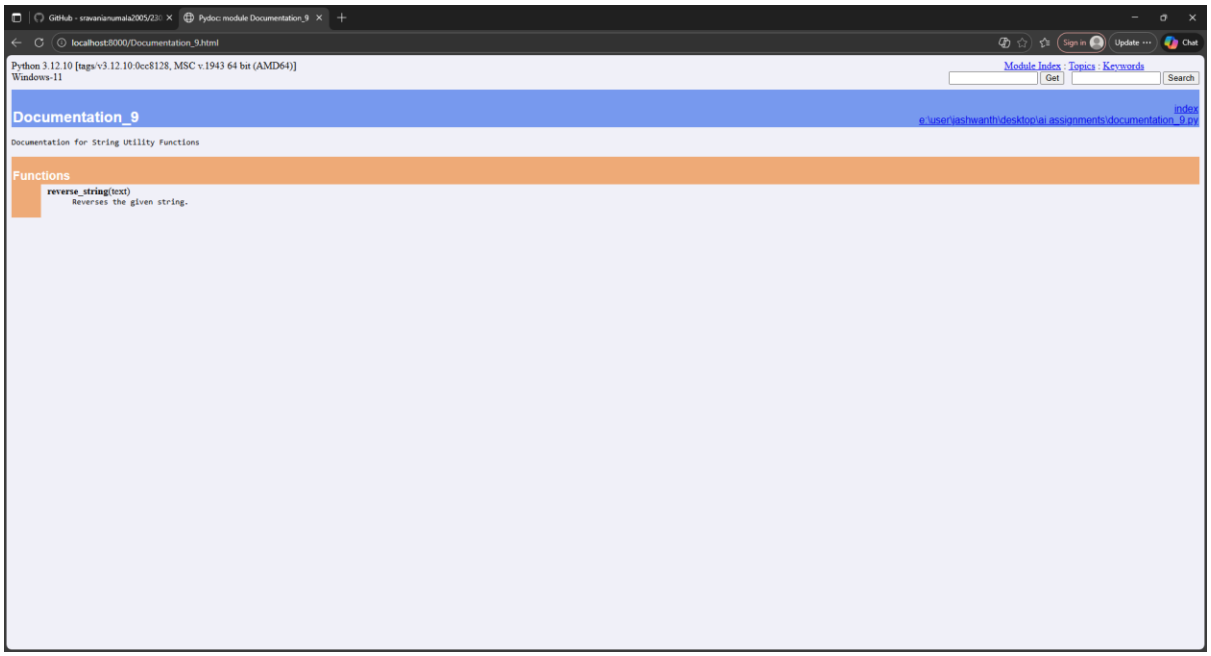
(c) Google-style documentation

CODE:




File Edit Selection View Go Run Terminal Help
AI ASSIGNMENTS
EXPLORER
AI ASSIGNMENTS
__pycache__
AI LAB TEST-1-10G7.py
AI-Assignment-1.5 10G7.pdf
AI-ASSIGNMENT-2.2-10G7.pdf
AI-ASSIGNMENT-2.5---10G7.pdf
AI-Assignment-3.1 10G7.pdf
AI-ASSIGNMENT-4.2-10G7.pdf
AI-ASSIgnment-4.5-10G7.pdf
AI-ASSIGNMENT-5.5-10G7.pdf
AI-ASSIGNMENT-6.5-10G7.pdf
AI-ASSIGNMENT-7.2-10G7.pdf
AI-ASSIGNMENT-8.2-10G7.pdf
Assignment-7.5.py
Assignment-8.2.py
Documentation_9.html
Documentation_9.py
Documentation_9.py
Documentation_9.html
Documentation_9.py > reverse_string
"""Documentation for String Utility Functions"""
def reverse_string(text):
"""Reverses the given string."""
return text[::-1]
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Documentation_9.py"
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> python Documentation_9.py
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> python -m pydoc -p 8000
Server ready at http://localhost:8000/
Server commands: [b]rowser, [q]uit
server> b
server>
Python
Python
OUTLINE
TIMELINE
Ln 5, Col 22
Spaces: 4
UTF-8
CRLF
Python
3.12.10 (Microsoft Store)


GitHub - sravanianumala2005/23
Pydoc: Index of Modules
localhost:8000
Summarize
Sign in
Update
Chat
Built-in Modules
_abc
_ast
_bisect
_blake2
_codecs
_codecs_cn
_codecs_hk
_codecs_iso2022
_codecs_jp
_codecs_kr
_codecs_tw
_collections
_contextvars
_csv
_datetime
_functools
_heapq
_imp
_io
_json
_locale
_lsprof
_md5
_multibytecodec
_opcode
_operator
_pickle
_random
_sha1
_sha2
_sha3
_signal
_sre
_stat
_statistics
_string
_struct
_symtable
_thread
_tokenize
_tracemalloc
_typing
_warnings
_weakref
_winapi
_xxinterpchannels
_xxsubinterpreters
array
atexit
audioop
binascii
builtins
cmath
errno
faulthandler
gc
itertools
marshal
math
mmap
msvcrt
nt
sys
time
unicodedata
winreg
xxsubtype
zlib
E:\User\Jashwanth\desktop\AI ASSIGNMENTS
AI LAB TEST-1-10G7
Documentation_9
C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.12_3.12.2800.0_x64__qbz5n2kfra8p0\python312.zip
C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.12_3.12.2800.0_x64__qbz5n2kfra8p0\DLLs
_asyncio
_bz2
_ctypes
_decimal
_elementtree
_hashlib
_lzma
_msi
_multiprocessing
_overlapped
_queue
_socket
_sqlite3
_ssl
_tkinter
_uuid
_wmi
_zoneinfo
pyexpat
select
unicodedata
winsound
C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.12_3.12.2800.0_x64__qbz5n2kfra8p0\Lib
__future__
__hello__
__phello__ (package)
_aix_support
_collections_abc
_compat_pickle
_compression
_markupbase
curses (package)
dataclasses
datetime
dbm (package)
decimal
difflib
dis
doctest
email (package)
multiprocessing (package)
netrc
nntplib
ntpath
nturl2path
numbers
opcode
operator
optparse
sre_constants
sre_parse
ssl
stat
statistics
string
stringprep
struct
subprocess
localhost:8000/Documentation_9.html



OBSERVATION:

The function reverse_string(text) is written multiple times with proper documentation strings explaining parameters and return values. In the terminal, the pydoc command is executed, and it successfully displays the function documentation. This shows that the docstring is correctly written and Python is able to generate documentation from it.

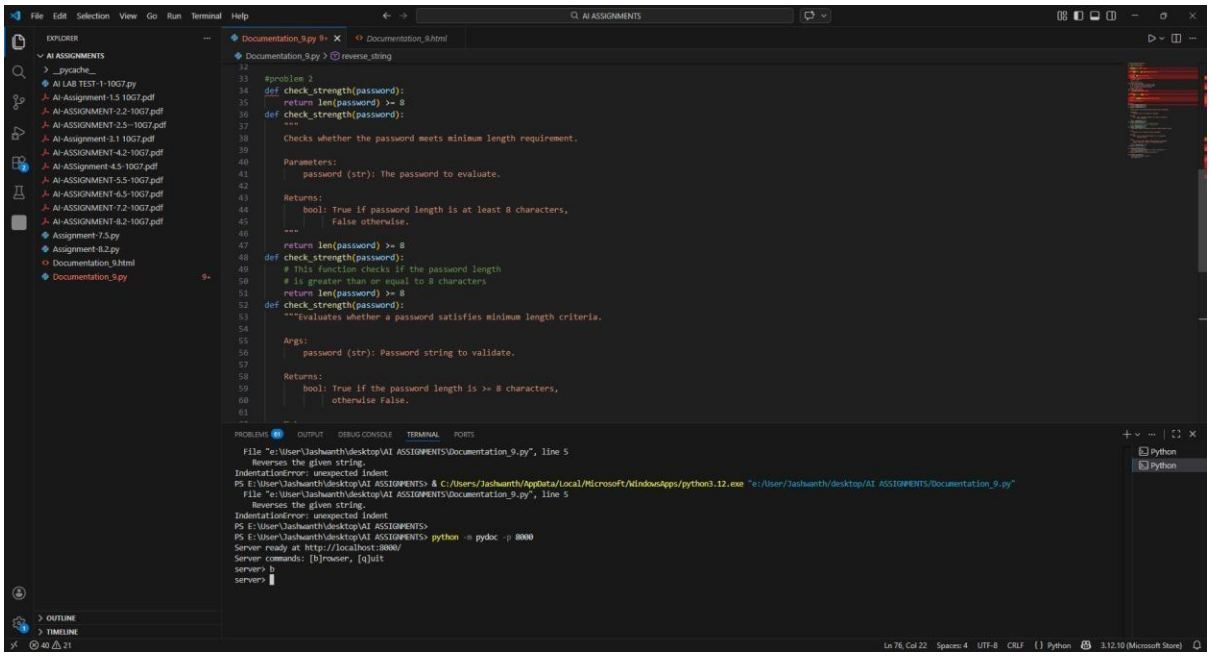
PROBLEM-2:

PROMPT:

def check_strength(password):

return len(password) >= 8

CODE:



OBSERVATION:

The function check_strength(password) includes structured Google-style documentation that clearly explains its purpose, input parameter, and return value. It specifies that the function checks whether the password length is at least 8 characters. The documentation contains sections like **Args** (describing the password as a string input) and **Returns** (indicating a boolean value: True if the password meets the minimum length requirement, otherwise False). This format improves clarity, readability, and maintainability of the code.

PROBLEM-3:

PROMPT: reate a module math_utils.py with functions:

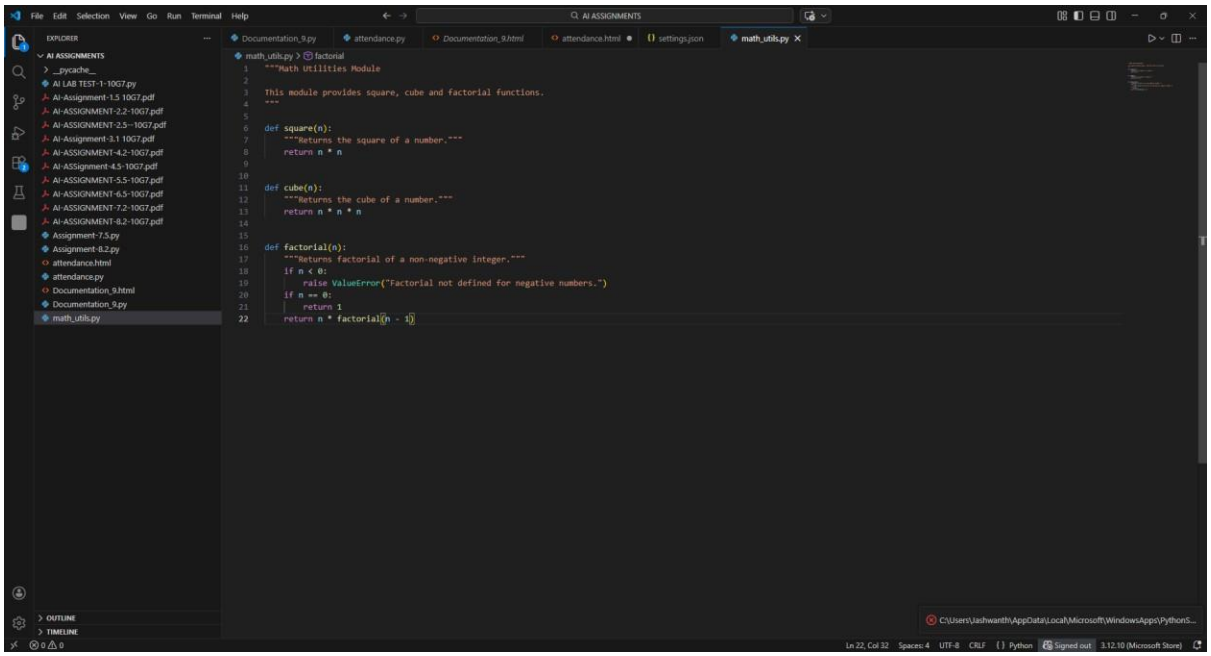
square(n)

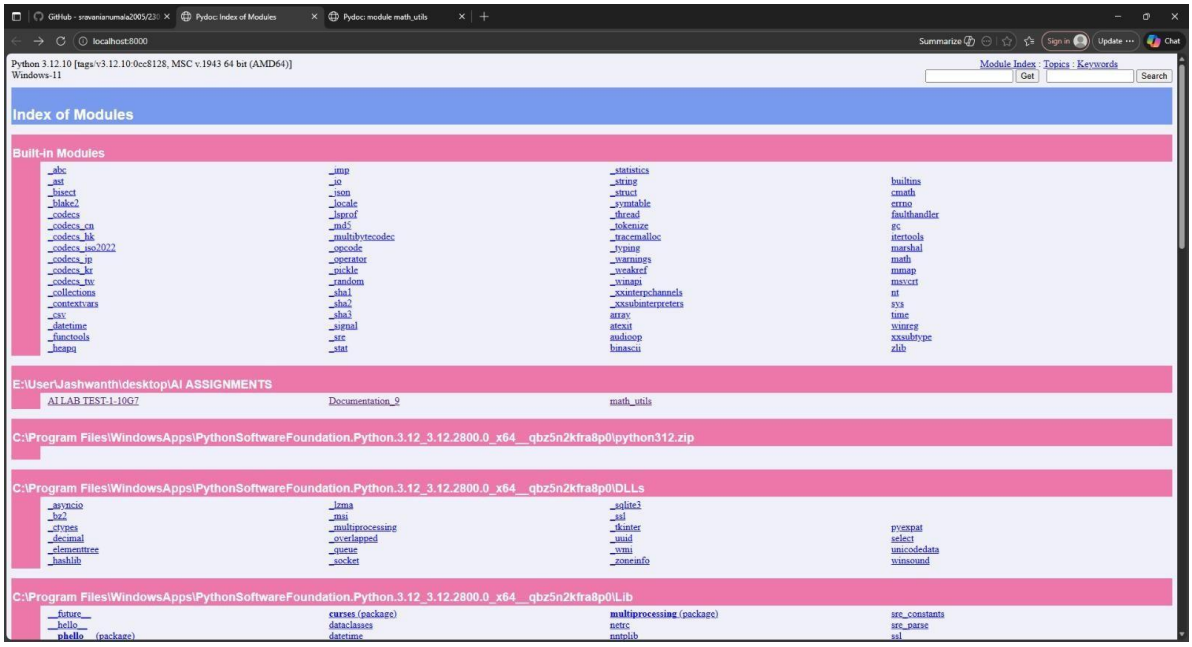
cube(n)

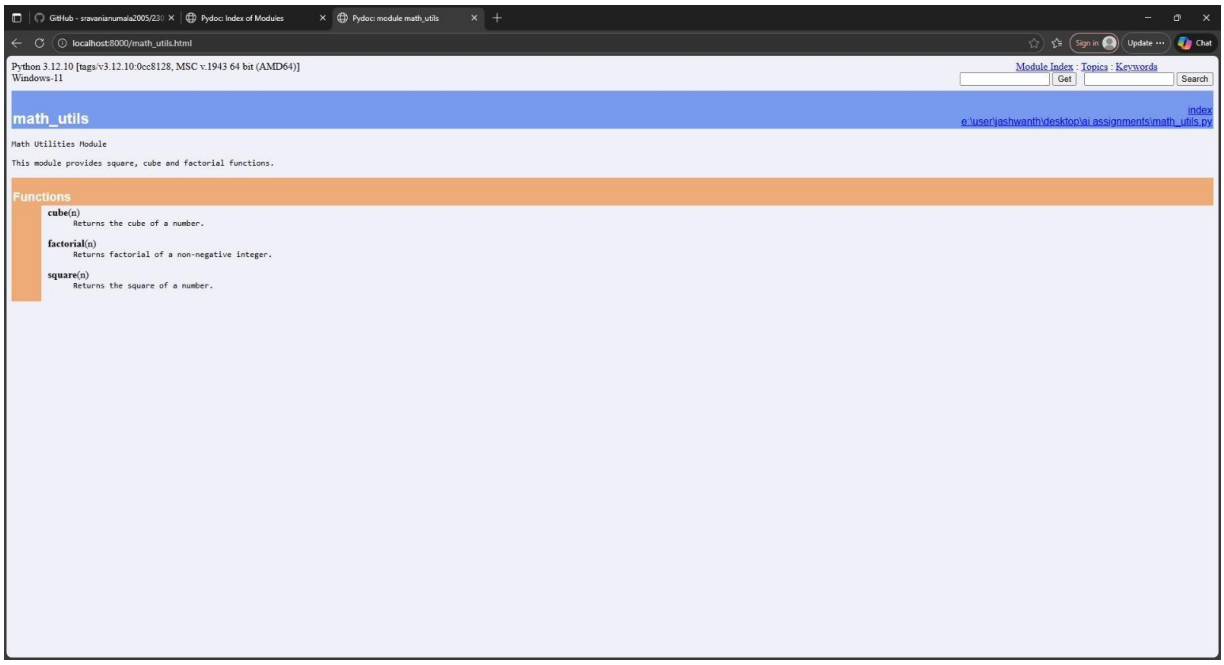
factorial(n)

Generate docstrings automatically using AI tools.

CODE:


File Edit Selection View Go Run Terminal Help
AI ASSIGNMENTS
EXPLORER
AI ASSIGNMENTS
__pycache__
AI LAB TEST-1-10G7.pdf
AI-Assignment-1.5 10G7.pdf
AI-ASSIGNMENT-2.2-10G7.pdf
AI-ASSIGNMENT-2.5---10G7.pdf
AI-Assignment-3.1 10G7.pdf
AI-ASSIGNMENT-4.2-10G7.pdf
AI-ASSIGNMENT-4.5-10G7.pdf
AI-ASSIGNMENT-5.5-10G7.pdf
AI-ASSIGNMENT-6.5-10G7.pdf
AI-ASSIGNMENT-7.2-10G7.pdf
AI-ASSIGNMENT-8.2-10G7.pdf
Assignment-7.5.py
Assignment-8.2.py
attendance.html
attendance.py
Documentation_9.html
Documentation_9.py
math_utils.py
Documentation_9.py
attendance.py
Documentation_9.html
attendance.html
settings.json
math_utils.py
math_utils.py > factorial
"""Math Utilities Module
This module provides square, cube and factorial functions.
"""
def square(n):
"""Returns the square of a number."""
return n * n
def cube(n):
"""Returns the cube of a number."""
return n * n * n
def factorial(n):
"""Returns factorial of a non-negative integer."""
if n < 0:
raise ValueError("Factorial not defined for negative numbers.")
if n == 0:
return 1
return n * factorial(n - 1)
OUTLINE
TIMELINE
C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\PythonS...
Ln 22, Col 32 Spaces: 4 UTF-8 CRLF { } Python Signed out 3.12.10 (Microsoft Store)


GitHub - sravanianumala2005/23
Pydoc: Index of Modules
Pydoc: module math_utils
localhost:8000
Summarize
Sign in
Update
Chat
Python 3.12.10 [tags:v3.12.10:0cc8128, MSC v.1943 64 bit (AMD64)]
Windows-11
Module Index : Topics : Keywords
Get
Search
Index of Modules
Built-in Modules
_abc _imp _statistics
_ast _io _string builtins
_bisect _json _struct cmath
_blake2 _locale _symtable errno
_codecs _lsprof _thread faulthandler
_codecs_cn _md5 _tokenize gc
_codecs_hk _multibytecodec _tracemalloc itertools
_codecs_iso2022 _opcode _typing marshal
_codecs_jp _operator _warnings math
_codecs_kr _pickle _weakref mmap
_codecs_tw _random _winapi msvcrt
_collections _sha1 _xxinterpchannels nt
_contextvars _sha2 _xxsubinterpreters sys
_csv _sha3 array time
_datetime _signal atexit winreg
_functools _sre audioop xxsubtype
_heapq _stat binascii zlib
E:\User\Jashwanth\desktop\AI ASSIGNMENTS
AI LAB TEST-1-10G7 Documentation_9 math_utils
C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.12_3.12.2800.0_x64__qbz5n2kfra8p0\python312.zip
C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.12_3.12.2800.0_x64__qbz5n2kfra8p0\DLLs
_asyncio _lzma _sqlite3
_bz2 _msi _ssl
_ctypes _multiprocessing _tkinter pyexpat
_decimal _overlapped _uuid select
_elementtree _queue _wmi unicodedata
_hashlib _socket _zoneinfo winsound
C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.12_3.12.2800.0_x64__qbz5n2kfra8p0\Lib
__future__ curses (package) multiprocessing (package) sre_constants
__hello__ dataclasses netrc sre_parse
__phello__ (package) datetime nntplib ssl



OBSERVATION:

The Python file contains three functions: square(n), cube(n), and factorial(n), each implemented with proper docstrings explaining their purpose.

Using pydoc, the documentation is automatically generated and displayed in the browser. The generated HTML page clearly lists the module name, a brief module description, and the available functions along with their explanations. The Index of Modules page also confirms that math_utils is correctly detected and included among the available modules.

This demonstrates that properly written docstrings enable automatic documentation generation, improving code readability and maintainability.

PROBLEM-4

PROMPT:

mark_present(student)

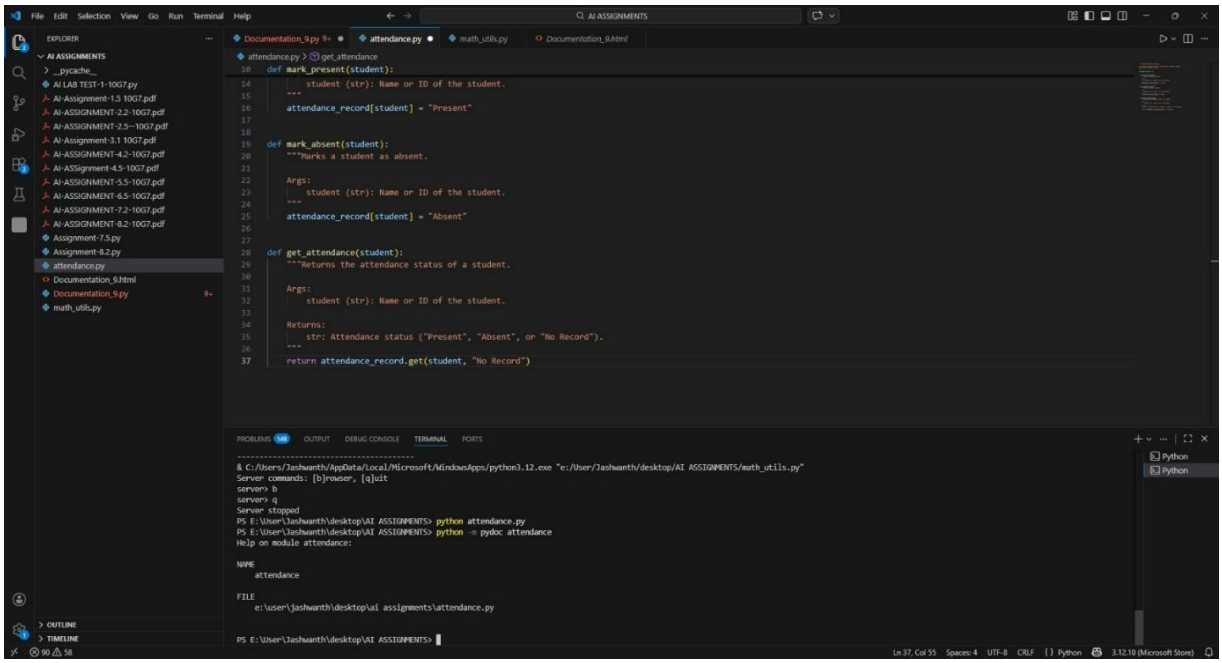
mark_absent(student)

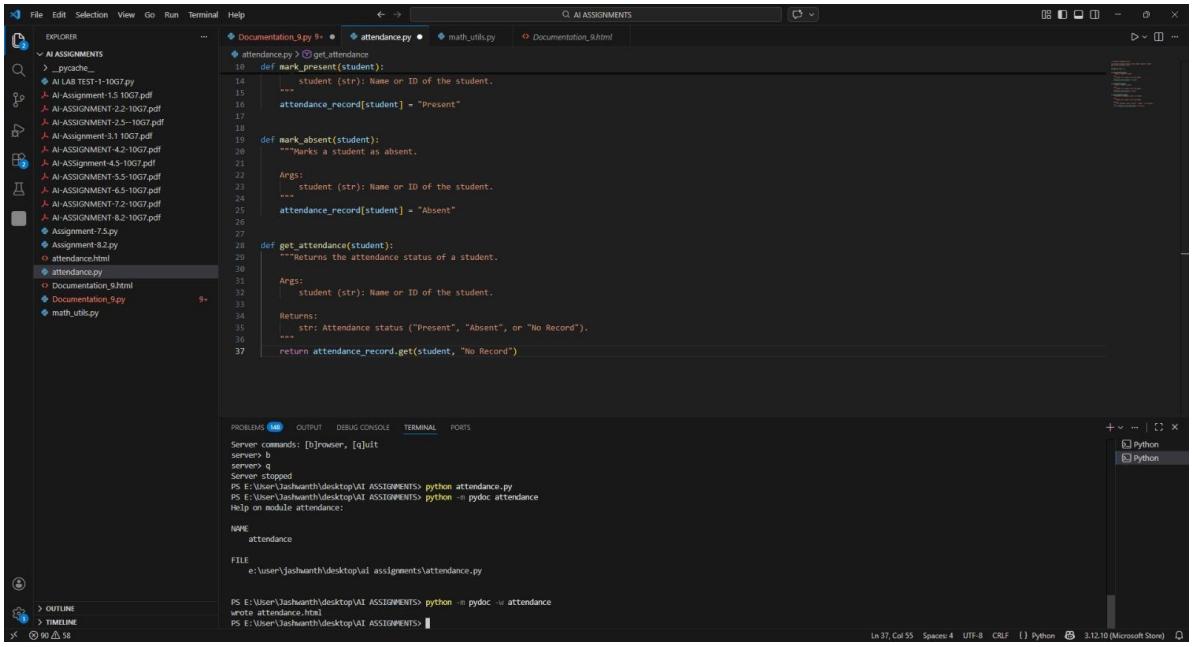
get_attendance(student)

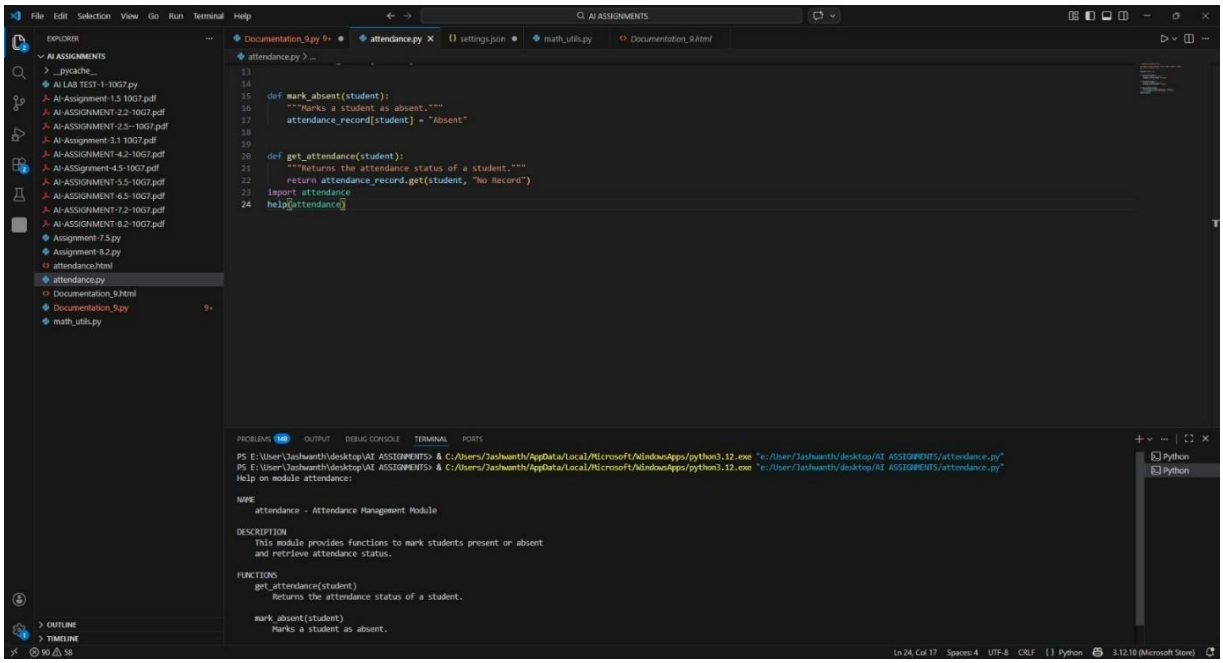
Add proper docstrings.

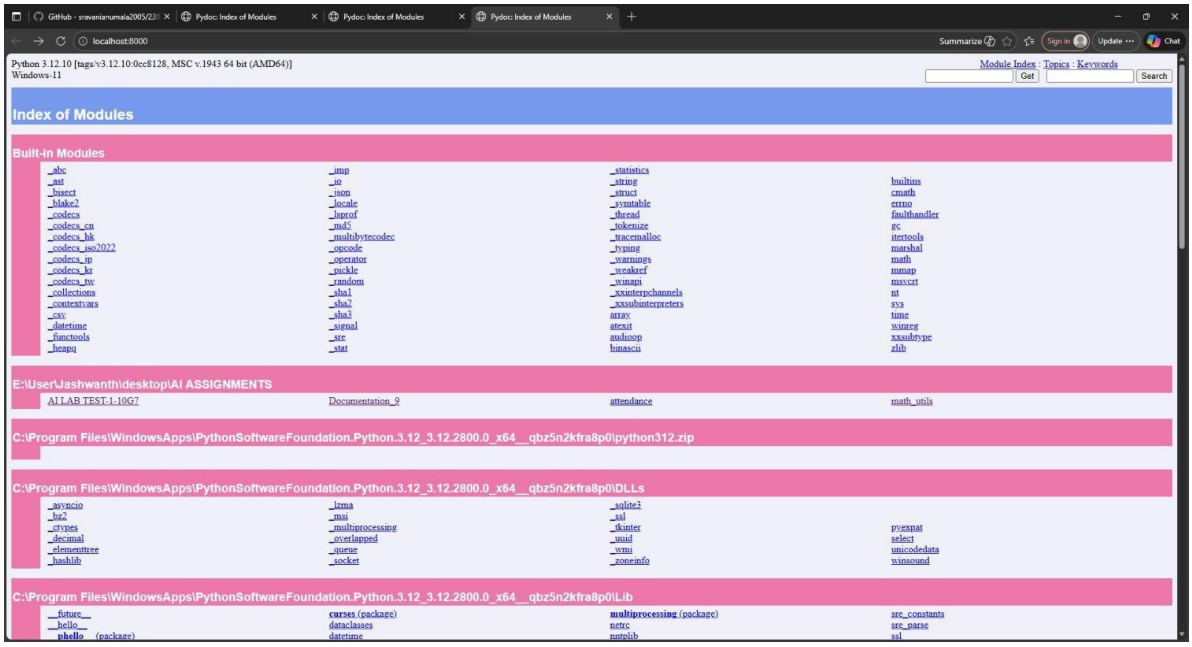
Generate and view documentation in terminal and browse

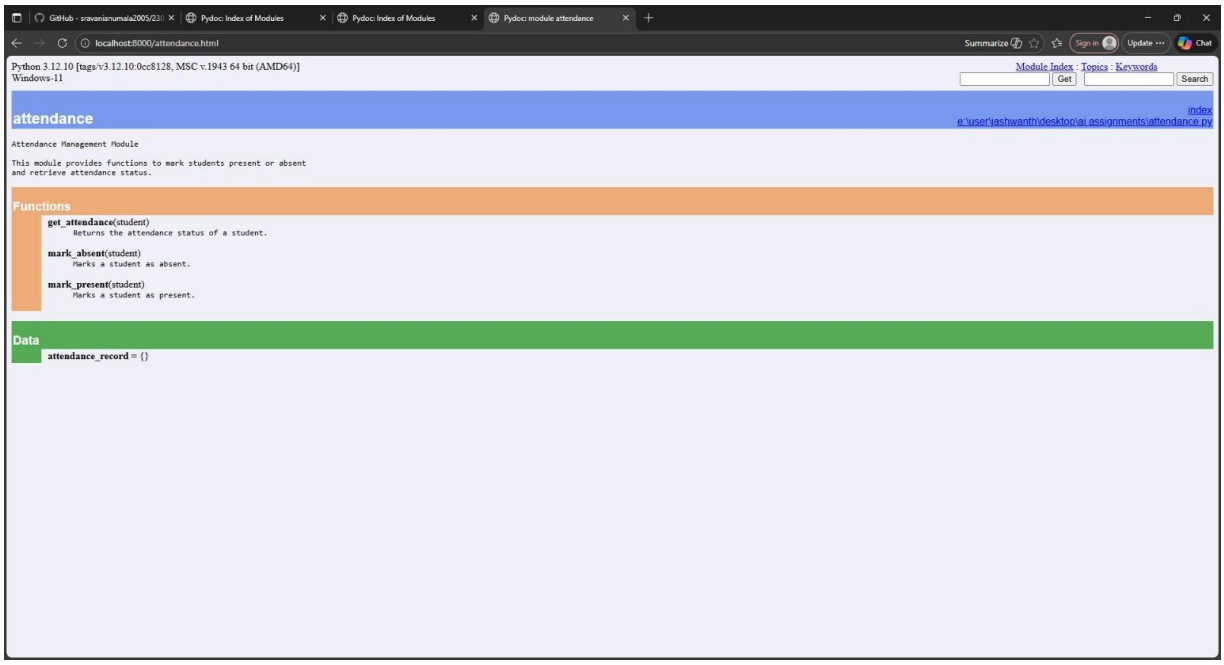
CODE:


File Edit Selection View Go Run Terminal Help
AI ASSIGNMENTS
EXPLORER
Documentation_9.py 9+
attendance.py
math_utils.py
Documentation_9.html
attendance.py > get_attendance
def mark_present(student):
student (str): Name or ID of the student.
attendance_record[student] = "Present"
def mark_absent(student):
"""Marks a student as absent.
Args:
student (str): Name or ID of the student.
attendance_record[student] = "Absent"
def get_attendance(student):
"""Returns the attendance status of a student.
Args:
student (str): Name or ID of the student.
Returns:
str: Attendance status ("Present", "Absent", or "No Record").
return attendance_record.get(student, "No Record")
PROBLEMS 548
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
& C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/math_utils.py"
Server commands: [b]rowser, [q]uit
server> b
server> q
Server stopped
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> python attendance.py
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> python -m pydoc attendance
Help on module attendance:
NAME
attendance
FILE
e:\user\jashwanth\desktop\ai assignments\attendance.py
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>
Python
Python
OUTLINE
TIMELINE
Ln 37, Col 55
Spaces: 4
UTF-8
CRLF
Python
3.12.10 (Microsoft Store)


File Edit Selection View Go Run Terminal Help
AI ASSIGNMENTS
EXPLORER
Documentation_9.py 9+
attendance.py
math_utils.py
Documentation_9.html
attendance.py > get_attendance
def mark_present(student):
student (str): Name or ID of the student.
attendance_record[student] = "Present"
def mark_absent(student):
"""Marks a student as absent.
Args:
student (str): Name or ID of the student.
attendance_record[student] = "Absent"
def get_attendance(student):
"""Returns the attendance status of a student.
Args:
student (str): Name or ID of the student.
Returns:
str: Attendance status ("Present", "Absent", or "No Record").
return attendance_record.get(student, "No Record")
PROBLEMS 548
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Server commands: [b]rowser, [q]uit
server> b
server> q
Server stopped
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> python attendance.py
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> python -m pydoc attendance
Help on module attendance:
NAME
attendance
FILE
e:\user\jashwanth\desktop\ai assignments\attendance.py
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> python -m pydoc -w attendance
wrote attendance.html
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>
Python
Python
OUTLINE
TIMELINE
Ln 37, Col 55
Spaces: 4
UTF-8
CRLF
Python
3.12.10 (Microsoft Store)


File Edit Selection View Go Run Terminal Help
AI ASSIGNMENTS
EXPLORER
Documentation_9.py 9+
attendance.py
settings.json
math_utils.py
Documentation_9.html
def mark_absent(student):
"""Marks a student as absent."""
attendance_record[student] = "Absent"
def get_attendance(student):
"""Returns the attendance status of a student."""
return attendance_record.get(student, "No Record")
import attendance
help(attendance)
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:/Users/Jashwanth/AppData/Local/Microsoft/WindowsApps/python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/attendance.py"
Help on module attendance:
NAME
attendance - Attendance Management Module
DESCRIPTION
This module provides functions to mark students present or absent
and retrieve attendance status.
FUNCTIONS
get_attendance(student)
Returns the attendance status of a student.
mark_absent(student)
Marks a student as absent.


Pydoc: Index of Modules
localhost:8000
Python 3.12.10 [tags:v3.12.10:0cc8128, MSC v.1943 64 bit (AMD64)]
Windows-11
Module Index : Topics : Keywords
Index of Modules
Built-in Modules
_abc _imp _statistics
_ast _io _string builtins
_bisect _json _struct cmath
_blake2 _locale _symtable errno
_codecs _lsprof _thread faulthandler
_codecs_cn _md5 _tokenize gc
_codecs_hk _multibytecodec _tracemalloc itertools
_codecs_iso2022 _opcode _typing marshal
_codecs_jp _operator _warnings math
_codecs_kr _pickle _weakref mmap
_codecs_tw _random _winapi msvcrt
_collections _sha1 _xxinterpchannels nt
_contextvars _sha2 _xxsubinterpreters sys
_csv _sha3 array time
_datetime _signal atexit winreg
_functools _sre audioop xxsubtype
_heapq _stat binascii zlib
E:\User\Jashwanth\desktop\AI ASSIGNMENTS
AI LAB TEST-1-10G7 Documentation_9 attendance math_utils
C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.12_3.12.2800.0_x64__qbz5n2kfra8p0\python312.zip
C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.12_3.12.2800.0_x64__qbz5n2kfra8p0\DLLs
_asyncio _lzma _sqlite3
_bz2 _msi _ssl
_ctypes _multiprocessing _tkinter pyexpat
_decimal _overlapped _uuid select
_elementtree _queue _wmi unicodedata
_hashlib _socket _zoneinfo winsound
C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.12_3.12.2800.0_x64__qbz5n2kfra8p0\Lib
__future__ curses (package) multiprocessing (package) sre_constants
__hello__ dataclasses netrc sre_parse
__phello__ (package) datetime nntplib ssl



OBSERVATION:

The module includes a clear title and a brief description stating that it manages student attendance. The documentation automatically lists the available functions such as get_attendance(student), mark_absent(student), and mark_present(student), along with short explanations of their purposes.

Additionally, the global data variable attendance_record = {} is displayed under the Data section, indicating that pydoc also documents module-level variables.

This confirms that properly written docstrings and structured code enable automatic and organized documentation generation using pydoc, improving clarity and maintainability of the module.

PROBLEM-5

PROMPT:

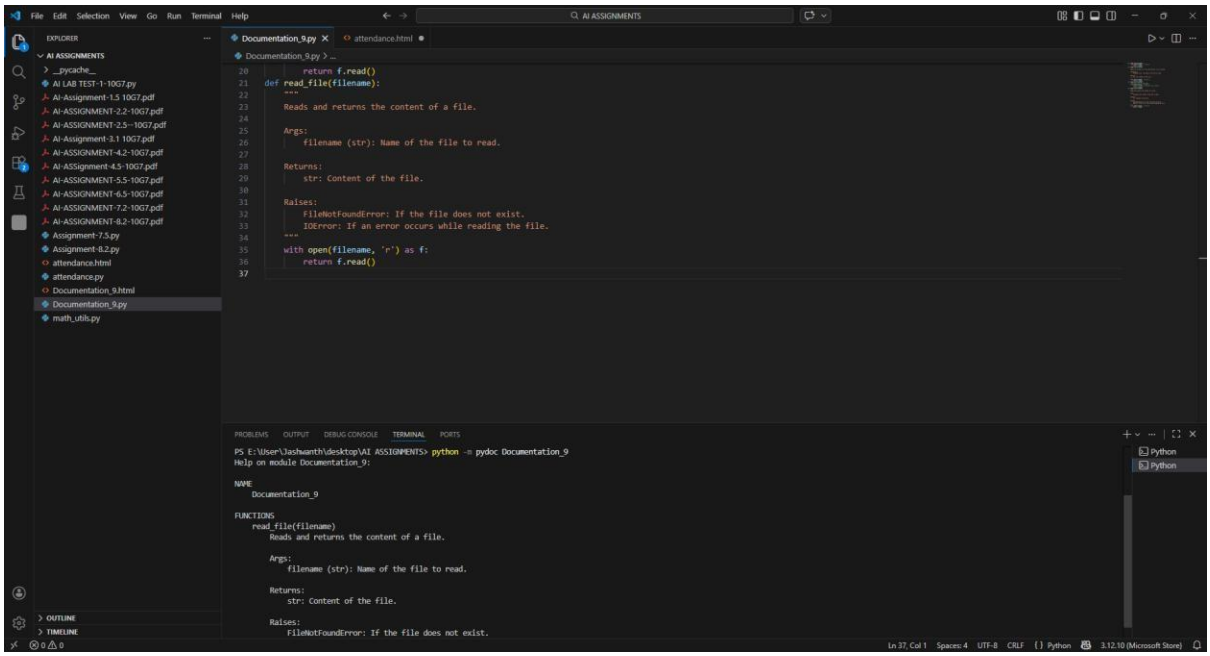
def read_file(filename):

with open(filename, 'r') as f:

return f.read()

Write documentation using all three formats.

CODE:



OBSERVATION:

The documentation was generated using pydoc, and the module details were displayed successfully in the terminal. This confirms that correctly written docstrings are recognized by Python.